Mining Visual Signatures of AI-Generated Images vs Real Images

Motivation:

AI-generated images are becoming increasingly common in online media, making it important to understand whether they still retain measurable visual signatures. This project asks whether simple, interpretable image features can distinguish AI-generated images from real images using a large public dataset and standard data mining methods. Rather than relying on deep neural networks, the analysis focuses on transparent visual statistics and their ability to reveal class differences in feature space and predictive modeling.

A **B**

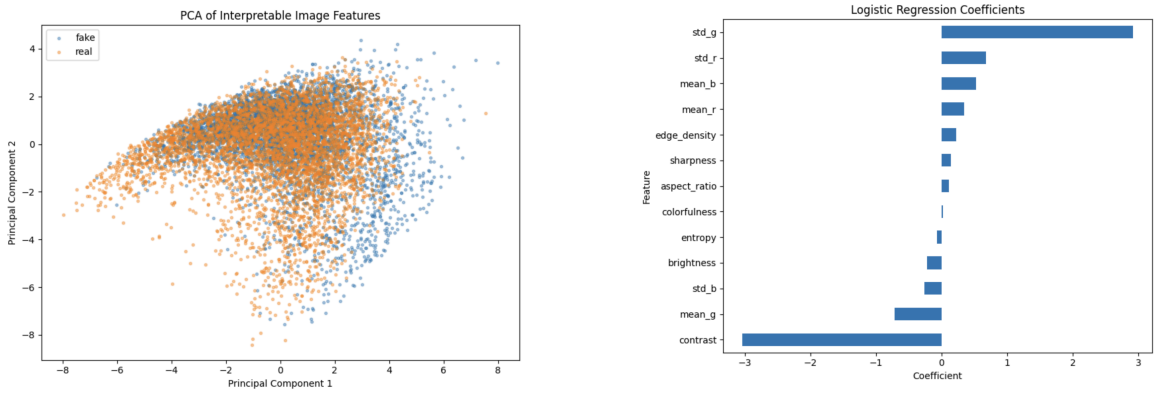

Figure 1. Visual separation and predictive feature importance for real vs AI-generated images. Panel A: Each point represents one image projected into two-dimensional PCA space using interpretable visual features. Colors indicate whether the image is **real** or **AI-generated**. **Panel B:** Bars show logistic regression coefficients for the same feature set. **Positive coefficients** indicate stronger association with **AI-generated images**, while **negative coefficients** indicate stronger association with **real images**. Larger absolute values indicate stronger influence on prediction. PCA and logistic regression were based on aspect ratio, RGB channel means and standard deviations, brightness, contrast, entropy, sharpness, edge density, and colorfulness.

Findings:

- The dataset contains **60,000 images**, balanced between **30,000 real** and **30,000 AI-generated** images, with balanced train and test splits.
- PCA shows substantial overlap between the two classes, but their centroids shift in opposite directions, indicating measurable differences in feature space. The first two principal components explain about 31.8% and 19.6% of total variance.
- AI-generated images tend to have higher average **sharpness**, **colorfulness**, **channel variability**, and **edge density** than real images in this dataset.
- A logistic regression model achieved about **0.639 accuracy** and **0.693 ROC-AUC**, showing that interpretable handcrafted features provide moderate but meaningful predictive power.

- Threshold analysis shows a strong tradeoff: at threshold **0.3**, recall is very high (**0.978**) but precision is low (**0.526**); at threshold **0.7**, precision improves (**0.721**) but recall drops sharply (**0.142**).
- Error analysis shows that **false positives** are unusually sharp, high-contrast, and edge-dense real images, while **false negatives** are visually softer fake images that more closely resemble natural photos.

Data and Methods:

This project uses a public image dataset containing **60,000 images (53 GB size)** organized into balanced **train (48,000) and test (12,000) splits**, with equal numbers of real and AI-generated images. The dataset includes both .jpg and .png files. Each image was resized to **256 × 256** and converted into a structured feature table using interpretable visual statistics, including aspect ratio, RGB channel means and standard deviations, brightness, contrast, Shannon entropy, Laplacian-based sharpness, Canny edge density, and colorfulness. The analysis combined descriptive feature comparison, **PCA** for low-dimensional visualization, and **logistic regression** for interpretable classification. **Model quality** was evaluated using accuracy, confusion matrix, ROC-AUC, threshold sensitivity, and error analysis on the held-out test split.

Model Behavior and Error Interpretation:

The model's performance is not only moderate overall, but also asymmetric in its mistakes. Real images that are falsely predicted as fake tend to be unusually sharp, high-contrast, edge-dense, and colorful, while fake images that are falsely predicted as real tend to be visually softer and more natural-looking. This suggests that the model is responding not only to authenticity, but also to broader image style characteristics such as visual intensity and texture.

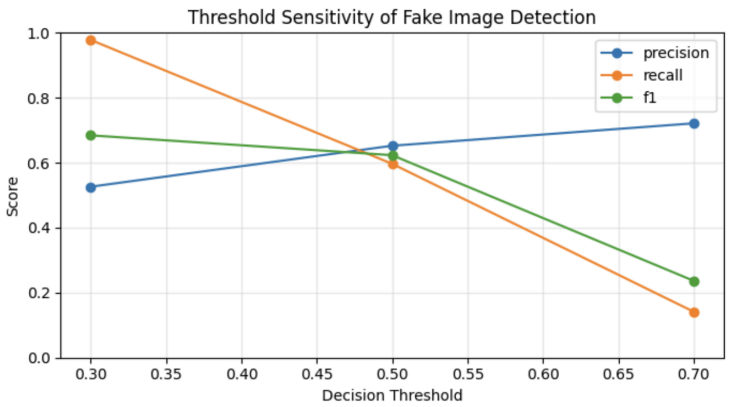

Figure 2. Threshold sensitivity of fake image detection.
Precision, recall, and F1-score change substantially as the decision threshold increases from 0.3 to 0.7. Lower thresholds maximize recall and identify most fake images, while higher thresholds improve precision but miss many fake images. This illustrates the tradeoff between sensitivity and reliability in simple interpretable detection models.

Significance Statement:

This project addresses an important topic because AI-generated images are becoming harder to identify visually, raising concerns about authenticity, trust, and misinformation. The analysis shows that even simple interpretable image features capture meaningful differences between real and synthetic images, but also reveals that these signals are only moderately predictive and can be confounded by broader visual style. This makes the project relevant both to media authenticity and to the limits of transparent, non-deep-learning detection methods.

Conclusion:

Overall, the project finds that AI-generated images do retain measurable visual signatures, especially in sharpness, edge density, and color variation, but these signatures do not fully separate fake images from real ones. PCA reveals a noticeable shift in feature space, while logistic regression shows that the features are moderately predictive but far from sufficient for highly reliable classification. Threshold and error analyses further show that model performance depends strongly on decision threshold and that visually intense real images can resemble synthetic ones, while softer fake images can resemble natural photos. Together, these findings suggest that interpretable handcrafted features are useful for exploratory detection, but limited as a standalone solution for robust authenticity verification.

GitHub Repository:

Repo Link: https://github.com/Janakrish1/Data_Mining_Project
Dataset: ai-generated-images-vs-real-images
This repository archives the Python notebooks, processed metadata and feature-generation workflow, figures, and final report materials.